

Design Thinking Lab Report -4

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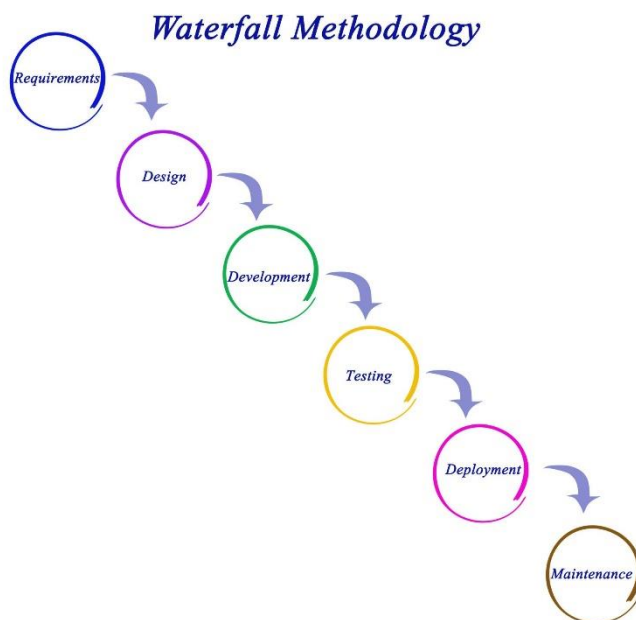
Topic: Result Processing System

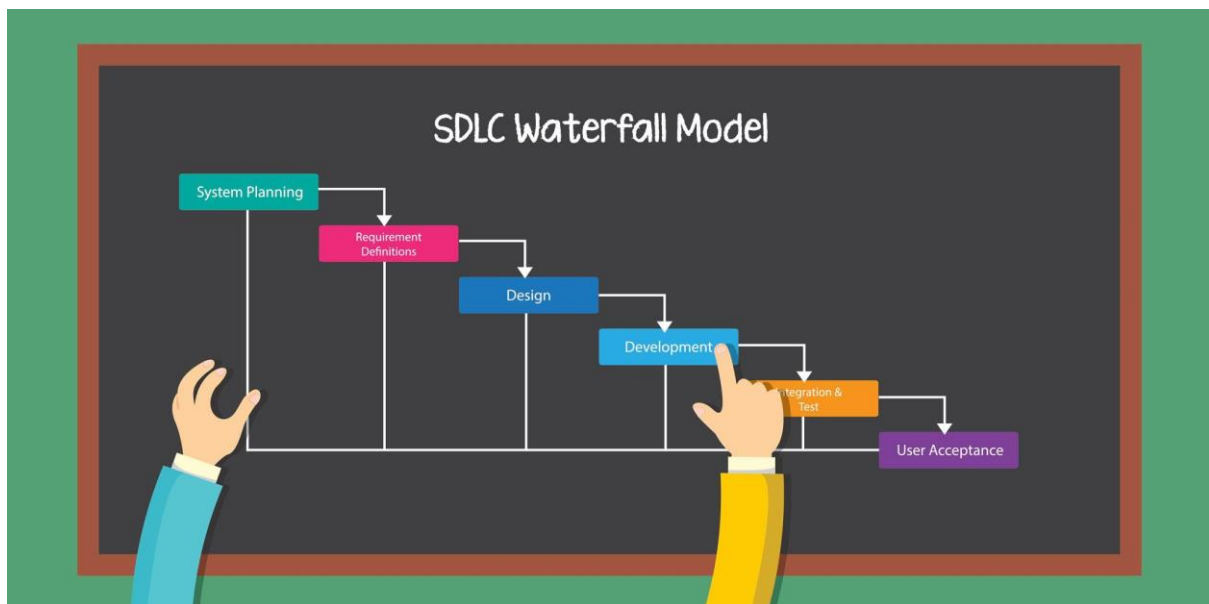
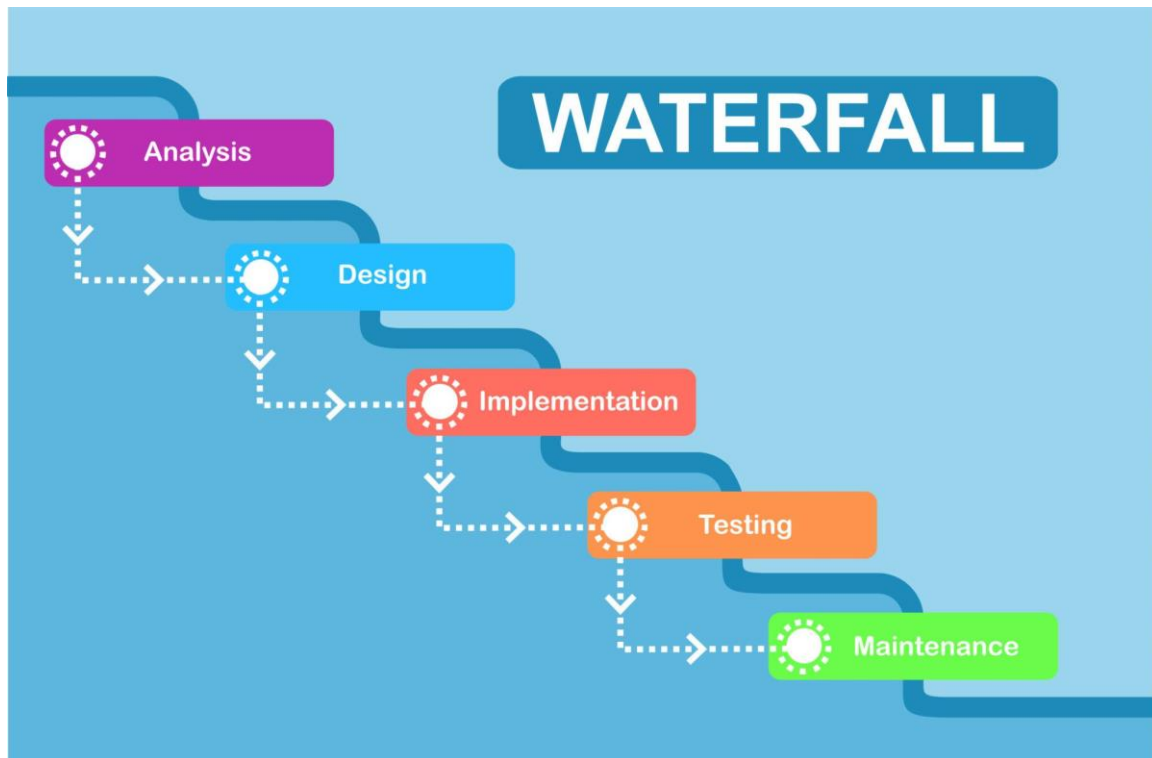
Aim of the Experiment

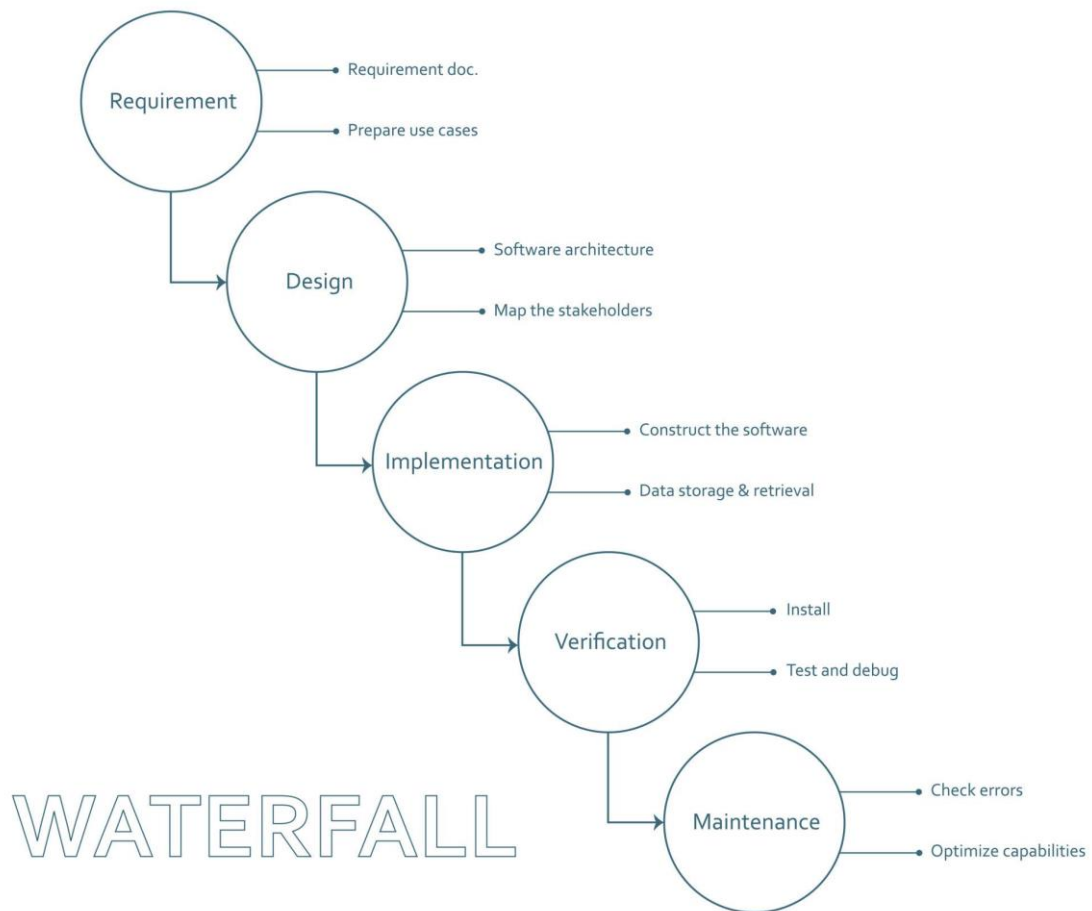
To analyze and compare the **Agile** and **Waterfall** software development models by applying them to a Smart Waste Management System and to study the project risks involved in developing such a system.

Overview of Development Models

1. Waterfall model





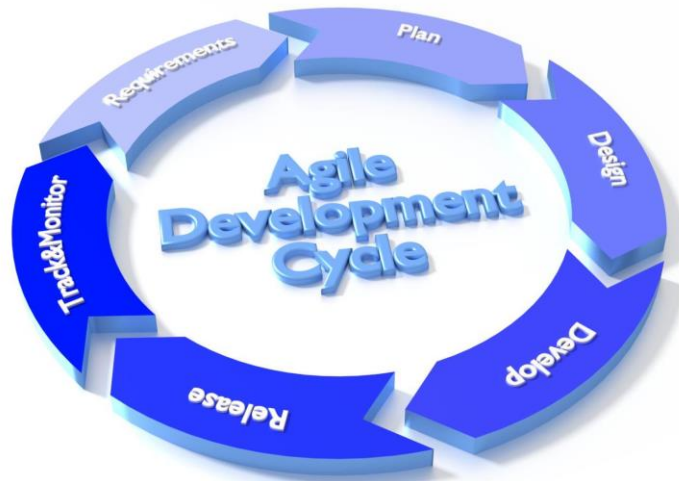


Characteristics:

- Linear and sequential process
- Each phase must be completed before the next begins
- Changes are difficult after requirement finalization

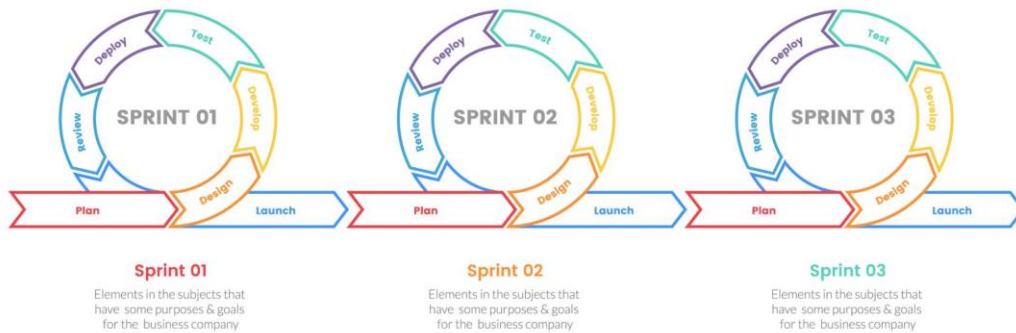
Phases:

1. Requirement Analysis
2. System Design
3. Implementation
4. Testing
5. Deployment & Maintenance



AGILE METHODOLOGY

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Characteristics:

- Iterative and incremental development
- Continuous customer feedback
- Flexible to requirement changes

Key Practices:

- Product backlog creation
- Sprint-based development
- Continuous integration and testing
- Regular stakeholder review

Application to Smart Waste Management System

Scenario Summary

The Smart Waste Management System includes:

- IoT-enabled garbage bins (fill-level sensors)

- Web dashboard for administrators
- Mobile app for collection staff
- Reporting and route optimization module

Requirements evolve over time due to:

- Environmental conditions
- Real-time data
- Feedback from municipal authorities

Application of Waterfall Model

Phase-wise Execution

1. Requirement Analysis

- All functional and non-functional requirements documented upfront.
- No flexibility for changes later.

2. System Design

- Complete architecture designed including:
 - Sensor integration
 - Database schema
 - Web & mobile UI design
 - Routing algorithm design

3. Implementation

- Coding strictly follows documented design.

4. Testing

- Testing begins after full system development.
- Sensor integration issues detected late.

5. Deployment

- Entire system deployed city-wide after completion.

Observations in Waterfall

- Difficult to add new sensor types later.
- Route optimization errors detected at final stage.
- High cost due to late defect detection.
- Risk of system failure after deployment.

Application of Agile Model

Agile-Based Execution

1. Product Backlog

- Initial features: bin monitoring, dashboard.
- Advanced features added progressively.

2. Sprint-wise Development

- Sprint 1: Real-time monitoring
- Sprint 2: Alerts + Mobile App
- Sprint 3: Route Optimization + Reports

3. Continuous Feedback

- Municipal authorities review after each sprint.
- Changes incorporated in next sprint.

4. Continuous Testing

- Sensor integration tested early.
- Performance issues detected quickly.

Observations in Agile

- Easy to handle changing requirements.
- Working system delivered early.
- Risk reduced due to iterative validation.
- Higher stakeholder involvement.

Risk Analysis

Risk Identified	Impact in Waterfall	Impact in Agile
Requirement Changes	High	Low
Sensor Integration Failure	Detected Late	Detected Early
Scalability Issues	High	Low
User Acceptance Risk	High	Low
Cost Overrun	High	Low

Comparative Analysis

Parameter	Waterfall	Agile
Requirement Flexibility	Low	High
Change Handling	Difficult	Easy
Risk Detection	Late	Early
Stakeholder Involvement	Minimal	Continuous
Suitability	Less Suitable	Highly Suitable

Result

The comparative study shows that the **Agile Model** is more suitable for developing the Smart Waste Management System because:

- Requirements evolve continuously.
- IoT integration requires early testing.
- Real-time data systems need iterative improvement.
- Risk detection is faster in Agile.

The Waterfall model is less suitable due to rigid structure and late-stage risk discovery.

Conclusion

For a Smart Waste Management System involving:

- IoT device integration
- Real-time monitoring
- Route optimization
- Continuous stakeholder feedback

The **Agile model** provides:

- Better flexibility
- Early risk mitigation
- Higher system quality
- Reduced development cost

Therefore, Agile is the recommended development methodology for this scenario.